MATH 243: Discussion Worksheet 10

Topics: 15.2 Double Integrals over General Regions; 15.3 Double Integrals in Polar Coordinates; 15.6 Triple Integrals
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1. Consider the double integral
$$\iint_D f(x,y)\,dA$$
where D is the region bounded by the lines $y = 3x + 1$, $x = 1$, and $y = 1$.
   (a) Sketch the region D.
   (b) Set up the double integral as an iterated integral in the order $dy\,dx$ (Type I).
   (c) Set up the double integral as an iterated integral in the order $dx\,dy$ (Type II).

2. Find the average value of the function $f(x,y) = y^2e^{xy}$ over the region D bounded by $y = x$, $y = 4$, and $x = 0$.

3. Consider the iterated integral
$$\int_0^1 \int_0^{y^2} f(x,y)\,dx\,dy.$$
   (a) Sketch the region of integration.
   (b) Reverse the order of integration.
   (c) Evaluate the integral in part (b) with the function $f(x,y) = \sqrt{x} + 3y$.

4. Using polar coordinates, find the volume of the solid that lies below the cone
$$z = \sqrt{x^2 + y^2}$$
and above the annulus $1 \le x^2 + y^2 \le 4$ in the xy-plane.

5. Sketch the region of integration. Then convert to polar coordinates and evaluate
$$\int_0^{1/2} \int_{\sqrt{3}\,y}^{\sqrt{1-y^2}} xy^2\,dx\,dy.$$

6. The following sum represents $\iint_D xy\,dA$ over a region D:
$$\int_{1/\sqrt{2}}^{1} \int_{\sqrt{1-x^2}}^{x} xy\,dy\,dx + \int_1^{\sqrt{2}} \int_0^x xy\,dy\,dx + \int_{\sqrt{2}}^{2} \int_0^{\sqrt{4-x^2}} xy\,dy\,dx.$$
Sketch D. Express the sum as a single iterated integral in polar coordinates, and evaluate it.

7. Evaluate each triple integral.
   (a) $\iiint_E (xy + z^2)\,dV$, where $E = [0,2] \times [0,1] \times [0,3]$.

(b) $\iiint_E \frac{1}{x^3}\,dV$, where $E = \{(x, y, z) \mid 0 \le y \le 1,\ 0 \le z \le y^2,\ 1 \le x \le z + 1\}$.

8. Set up and evaluate a triple integral to find the volume of the tetrahedron in the first octant bounded by the coordinate planes and the plane passing through $(2, 0, 0)$, $(0, 4, 0)$, and $(0, 0, 12)$.

SOME USEFUL DEFINITIONS, THEOREMS, AND NOTATION

Double Integrals over General Regions. Suppose that f is continuous on a region D. Assume that the boundary functions below are continuous on their respective intervals.

- A Type I region has the form $D = \{(x, y) \mid a \le x \le b,\ g_1(x) \le y \le g_2(x)\}$. Then

$$\iint_D f(x, y)\,dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y)\,dy\,dx.$$

- A Type II region has the form $D = \{(x, y) \mid c \le y \le d,\ h_1(y) \le x \le h_2(y)\}$. Then

$$\iint_D f(x, y)\,dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y)\,dx\,dy.$$

Double Integrals in Polar Coordinates. Suppose that f is continuous on a region D in the plane, and that D is described in polar coordinates by

$$\alpha \le \theta \le \beta, \qquad h_1(\theta) \le r \le h_2(\theta),$$

where h_1 and h_2 are continuous on $[\alpha, \beta]$, $0 \le h_1(\theta) \le h_2(\theta)$, and $0 < \beta - \alpha \le 2\pi$. Then

$$\iint_D f(x, y)\,dA = \int_\alpha^\beta \int_{h_1(\theta)}^{h_2(\theta)} f(r\cos\theta, r\sin\theta)\,r\,dr\,d\theta.$$

Triple Integrals. Suppose that f is continuous on a solid

$$E = \{(x, y, z) \mid (x, y) \in D,\ u_1(x, y) \le z \le u_2(x, y)\},$$

where D is a Type I or Type II region as above, and u_1, u_2 are continuous on D. Then

$$\iiint_E f(x, y, z)\,dV = \iint_D \left(\int_{u_1(x,y)}^{u_2(x,y)} f(x, y, z)\,dz \right) dA,$$

and we use a Type I or Type II description of D to write this as an iterated integral. Other orders of integration are handled similarly. The volume of E is $V = \iiint_E 1\,dV$.

Suggested Textbook Problems

Section 15.2: 1-50, 55-79 Section 15.3: 1-49 Section 15.6: 1-47, 57, 58